

MODULE 1.2: Research Formulation & Design

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Literature Review

A **literature review** is a systematic examination and evaluation of existing scholarly work related to a specific research topic. It involves collecting, analysing, summarizing, and synthesizing previous studies to understand the current state of knowledge and identify gaps for further research. **Steps:**

- i. **Define the Research Problem** – Identify the key topic and scope of the review and choose the research topic precisely.
- ii. **Search for Relevant Literature** – Use academic databases (e.g., Google Scholar, IEEE Xplore, PubMed) to access relevant literature like journals, articles, paper works.
- iii. **Evaluate and Select Sources** – Choose credible and relevant studies, as studying everything written so far is impossible so selection of relevant ones is important.
- iv. **Analyse and Organize Information** – note down common findings, important trends or conclusions, influential theories, major themes and patterns.
- v. **Identify Research Gaps** – Find areas that need further investigation.
- vi. **Outline the structure of literature review:** different approaches are available such as:
 1. Chronological (latest one written at last)
 2. Thematic (similar one written together)
 3. Methodological and theoretical
- vii. **Write LR** – Write a well-structured review that connects past studies to the current research.
 1. Should be clear and concise
 2. Should be logical
 3. No grammatical or spelling mistakes
 4. Avoid plagiarism and use citation.

Purpose of Literature Review

- **Provides Background Information:** Helps researchers understand existing knowledge in the field.
- **Identifies Research Gaps:** Highlights areas that require further investigation.
- **Avoids Duplication:** Ensures that new research contributes uniquely rather than repeating past studies.
- **Guides Research Methodology:** Assists in selecting appropriate methods based on previous studies.
- **Justifies Research Significance:** Demonstrates the importance of the study in the broader research context.
- **Establishes a Theoretical Framework:** Connects the study to existing theories.

- **Helps in Forming Hypotheses:** Provides a basis for developing research hypotheses.
- **What has been discovered so far?**
- **Identify conflicts in previous studies.**

Primary and Secondary Sources of Literature review

1. **Primary Sources:** Primary sources are original, first-hand accounts of research, events, or discoveries. These sources provide direct evidence and have not been interpreted or analyzed by others. **Types of Primary Sources**
 - **Research Articles** – Original research studies published in academic journals.
 - **Theses and Dissertations** – Research work submitted by students for academic degrees.
 - **Conference Papers** – Papers presented at conferences, containing recent research.
 - **Patents** – Official documents granting rights for inventions or discoveries.
 - **Government Reports** – Official documents detailing research findings or policies.
 - **Interviews and Surveys** – First-hand data collected through direct interaction.
 - **Laboratory Data and Experiments** – Raw data obtained from scientific experiments.
 - **Manuscripts and Historical Documents** – Handwritten or unpublished records.
2. **Secondary Sources:** interpret, analyze, or summarize primary sources. They provide an overview or evaluation of existing research. **Types of Secondary Sources**
 - **Review Articles** – Summaries of multiple research papers, providing an analysis of trends.
 - **Textbooks** – Books that compile and explain knowledge on a subject.
 - **Encyclopedias & Dictionaries** – Reference materials summarizing knowledge in a field.
 - **Systematic and Critical Reviews** – Structured analyses of research.
 - **Biographies** – Accounts of a person's life based on primary sources.
 - **Commentaries and Editorials** – Expert opinions on published research.
 - **Meta-Analyses** – Statistical studies combining results from multiple primary sources.

Reviews

Reviews in research help summarize, analyze, and synthesize existing literature to provide insights into a topic. There are different types of reviews, each serving a specific purpose.

1. Narrative Review

- A **descriptive summary** of literature on a broad topic.
- No formal process or criteria—**author selects sources** based on relevance.
- Useful for getting an **overview of a field**, identifying themes, and building background knowledge.
- Example: A review of how AI has evolved in healthcare over the past decade.

2. Systematic Review

- A **highly structured** review that follows a step-by-step process to identify, evaluate, and synthesize studies on a **specific question**. [well defined more rigorous approach](#)
- Involves pre-defined **inclusion/exclusion criteria**, **quality assessment**.
- Minimizes bias and ensures reproducibility.
- Example: Reviewing all studies on the effectiveness of online education during COVID-19.

3. Literature Review

- A **broader and more general** form of review often included in **thesis or research papers**.
- Aims to **summarize past research**, identify gaps, and establish a theoretical base for new work.
- Less rigid than systematic reviews but more structured than a narrative review.

4. Meta-analysis

- A **statistical technique** used to combine data from multiple studies. [about the same topic/subject](#)
- Provides a **quantitative summary**—like a pooled average or effect size. [used deductive approach/top down approach](#)

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5. Meta synthesis: is based on non statistical technique, that integrated,evaluates and interpretes findings of multiple quanlitative studies. it is inductive research approach

- Always follows a **systematic review** and adds numerical insights.
- Example: Combining results from 10 studies to determine the average impact of exercise on stress.

5. Critical Review

- Goes **beyond description**—it evaluates and critiques the existing literature.
- Focuses on **strengths, weaknesses, assumptions**, and theoretical gaps.
- Often used to **challenge existing viewpoints or propose new theories**.

6. Integrative Review

- Brings together both **theoretical and empirical** work—quantitative and qualitative.
- Aims to create **new frameworks, models, or perspectives** from the mix.
- Often used in fields like education, nursing, or management.

7. Rapid Review

- A **time-saving approach** to a systematic review, often used in urgent policy or clinical decisions.
- Involves **streamlining steps** like fewer databases or simplified quality checks.
- Example: Quickly assessing which COVID-19 treatments showed early promise.

Monographs and Patents

1. **Monograph** is a detailed, single-authored study on a specific single topic or subject. It provides in-depth knowledge, often focusing on academic or research topics. **Features:**
 - Covers a single subject extensively.
 - Based on original research or detailed analysis.
 - These are reports published on original research that are too long and specialized.
 - Often published by universities or research institutions.

Importance of Monographs in Research

- Provides comprehensive insights into a specialized topic.
- Serves as a reliable academic reference.
- Used as **secondary** literature in research.

2. **Patents:** is an official document that grants an inventor exclusive rights to their invention for a specific period. It ensures that the inventor can control the commercial use of their innovation. **Types of Patents**
 - **Utility Patents** – Protect functional inventions, such as new machines or processes.
 - **Design Patents** – Cover ornamental designs of functional items.
 - **Plant Patents** – Protect new plant varieties.

Importance of Patents in Research

- Encourages innovation by protecting intellectual property.
- Provides technical details about inventions for future research.
- Helps industries develop new products based on patented technologies.
- Prevents plagiarism

Research Databases

Research databases are online repositories that provide access to scholarly articles, journals, theses, and patents. **Common Research Databases**

- **Google Scholar** – Free access to a wide range of academic papers.
- **IEEE Xplore** – Focuses on engineering and technology research.
- **PubMed** – Covers medical and life sciences research.
- **ScienceDirect** – Provides access to scientific, technical, and medical journals.
- **JSTOR** – Offers academic journal articles in humanities, social sciences, and sciences.

- **SpringerLink** – Contains journals, books, and conference papers.
- **Scopus** – A large citation database with peer-reviewed literature.
- **Web of Science** – Indexes high-impact scientific journals and conference proceedings.

Web as a Source in Research

1. Importance of the Web for Research: The web serves as a vast and dynamic resource for researchers, providing access to scholarly articles, reports, and open-access databases. **Benefits:**

- **Easy Accessibility** – Available anytime and anywhere.
- **Diverse Sources** – Provides academic papers, government reports, blogs, and discussion forums.
- **Real-Time Information** – Updated frequently with the latest research.
- **Open-Access Journals** – Enables access to research without paywalls.

2. Tools for Web-Based Research

Search Engines & Research Databases:

- **Google Scholar** – Academic papers, citations, and open-access journals.
- **PubMed** – Medical and life sciences literature.
- **IEEE Xplore** – Engineering and technology-related research.
- **ScienceDirect** – Peer-reviewed journal articles.
- **JSTOR** – Humanities and social sciences publications.

Reference Management Tools:

- **Zotero** – Organizes references and citations.
- **Mendeley** – Manages PDFs and collaborative research.
- **EndNote** – Citation management tool for scholarly writing.

AI & Data Analysis Tools:

- **ChatGPT & Elicit** – Assists in summarizing and generating research insights.
- **Google Bard** – Web-based AI for research queries.

3. Web Search Techniques

Boolean Operators:

- **AND** – Finds results containing all specified terms (e.g., "Machine Learning AND Healthcare").
- **OR** – Includes results with any of the specified terms (e.g., "AI OR Deep Learning").
- **NOT** – Excludes terms from results (e.g., "Climate Change NOT Politics").

Advanced Search Methods:

- **Phrase Searching** – Use quotation marks for exact matches (e.g., "Artificial Intelligence applications").
- **Truncation & Wildcards** – Use * or ? for variations (e.g., "comput*" finds "computer" or "computing").
- **Site-Specific Search** – Use **site:** to search within a website (e.g., "Quantum Computing site:nasa.gov").

4. Using Web Tools for Research

- **Google Alerts** – Tracks new research on specific topics.
- **ResearchGate & Academia.edu** – Platforms for sharing and discovering academic papers.
- **Semantic Scholar** – AI-powered research paper search engine.

Biases in Literature

Bias in literature refers to systematic errors or distortions in research due to various factors such as selective reporting, researcher preferences leading to an inaccurate representation of the truth in research findings.

- Understanding these biases is essential for conducting an objective and critical literature review.
- Bias can affect various stages of research from design and data collections to analysis and interpretation.
- **Types of Bias in Literature**

1. **Selection Bias:** Occurs when the sample is not the representation of the population being studied. Certain studies or data are included/excluded in a way that skews results. **Example:** A medical review on drug effectiveness only on young healthy adults may not provide accurate result for older/ill population.
2. **Publication Bias:** Favoring studies with significant or positive results over those with neutral/negative results. **Example:** Journals are more likely to publish studies showing a new treatment works rather than studies that show no effect.
3. **Confirmation Bias:** Researchers selectively use sources that support their pre-existing beliefs while ignoring contradictory evidence. **Example:** A study on climate change only cites sources that confirm human-caused warming and ignores other perspectives.
4. **Response Bias:** When participants give inaccurate or socially desirable answers rather than truthful ones. **Example:** In a survey about alcohol consumption, people may underreport how much they actually drink.
5. **Measurement Bias:** Occurs when the tools or methods used to collect data consistently misrepresent what they are supposed to measure. **Example:** A faulty weighing scale always showing 2 kg less than the actual weight.
6. **Recall Bias:** Happens when participants don't remember past events accurately, affecting data quality. **Example:** A patient may forget or misreport the exact dosage of a medicine they took a year ago.
7. **Sampling Bias:** Occurs when the sample is not representative of the population being studied. **Example:** Conducting an income survey by only sampling people at luxury malls.
8. **Language Bias:** Studies published in certain languages are preferred over others, leading to incomplete literature coverage. **Example:** A researcher only includes English-language studies in a review, ignoring relevant studies in other languages.
9. **Funding Bias:** Research outcomes are influenced by the interests of the funding organization. **Example:** A study funded by a tobacco company finds no clear link between smoking and cancer.
10. **Time-Lag Bias:** Delayed publication of negative or insignificant results compared to positive results. **Example:** A pharmaceutical company delays publishing a study showing that its new drug has serious side effects.

A. Strategies to Identify Bias

- **Diversify Sources** – Include studies from different journals, disciplines, and geographical locations.
- **Compare Conflicting Results** – Analyze studies with contradictory findings to assess broader perspectives.

- **Check Methodologies** – Evaluate research design, sample size, and data collection methods.
- **Analyze Funding Sources** – Identify potential conflicts of interest.
- **Assess Citation Patterns** – Look for over-reliance on specific sources or theories.
- **Use Open-Access Databases** – Ensure access to a wider range of research, not just paywalled journals.
- **Acknowledge Limitations** – Clearly state any biases or gaps in the review.
- **Use Meta-Analysis** – Combine multiple studies to get a more balanced view.